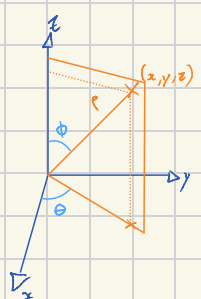
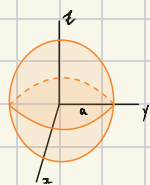


Lecture 6: Spherical coordinates, surface area



$\rho = \lambda r = \text{distance from the origin}$
 $\phi = \text{angle down from } z\text{-axis } (0 \leq \phi \leq \pi)$
 $\theta = \text{angle from } x\text{-axis } (0 \leq \theta \leq 2\pi)$

On a sphere $\rho = a$



$\phi \leftrightarrow \text{latitude}$
 $\theta \leftrightarrow \text{longitude}$



$\lambda = \text{distance from } z\text{-axis}$
 "polar coordinates in xz-plane"
 $z = \rho \cos \phi$
 $\lambda = \rho \sin \phi$

$$x = \lambda \cos \theta = \rho \sin \phi \cos \theta$$

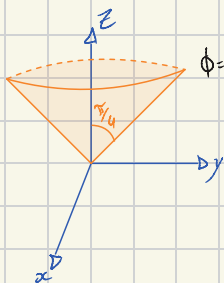
$$y = \lambda \sin \theta = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

$$\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{x^2 + y^2 + z^2}$$

• $\rho = a$: sphere of radius a centered at O .

• $\phi = \frac{\pi}{4}$ is a cone

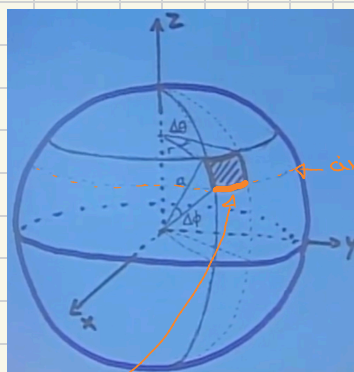


$\phi = \frac{\pi}{4}; z = \lambda$

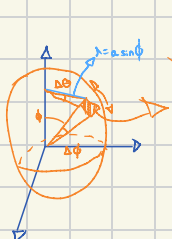
$\phi = \frac{\pi}{2}$ is the xy -plane

Triple Integrals in spherical coordinates: Volume component? $dV \approx \Delta \rho \Delta \phi \Delta \theta$

$$\Delta V \approx \Delta \rho \Delta \phi \Delta \theta$$



piece of circle of radius a , length $a \Delta \phi$

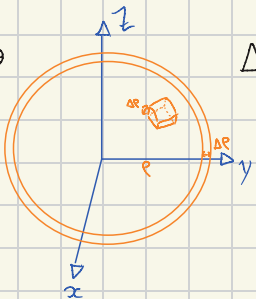


piece of circle of radius λ , length $\lambda \Delta \theta = a \sin \phi \Delta \theta$

Surface area on sphere of radius a ?

$$\Delta S \approx (a \sin \phi \Delta \theta) \cdot (a \Delta \phi) = a^2 \sin \phi \Delta \phi \Delta \theta$$

$$dS = a^2 \sin \phi d\phi d\theta$$

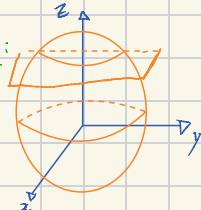


$$\Delta V \approx \Delta \rho \cdot \Delta S$$

$$= \rho^2 \sin \phi \Delta \rho \Delta \phi \Delta \theta$$

$$dV = \rho^2 \sin \phi d\rho d\phi d\theta$$

Example:

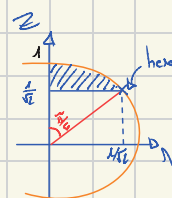


vol. of portion of unit sphere above $z = \frac{1}{\sqrt{2}}$.

$$\int_0^{2\pi} \int_0^{\pi/4} \int_{1/\sqrt{2}}^1 \rho^2 \sin \phi d\rho d\phi d\theta$$

Evaluation: $\frac{2\pi}{3} - \frac{5\pi}{16\sqrt{2}}$

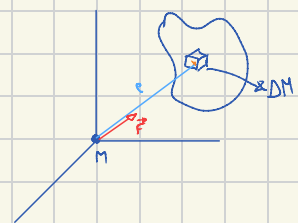
Plane: $z = \frac{1}{\sqrt{2}}$
 $\rho \cos \phi = \frac{1}{\sqrt{2}}$
 $\rho = \frac{1}{\sqrt{2} \cos \phi}$



here: $\rho = 1$ (on the sphere) and $\rho \cos \phi = \frac{1}{\sqrt{2}}$ so $\cos \phi = \frac{1}{\sqrt{2}} = \phi = \frac{\pi}{4}$

Applications: gravitational attraction

Gravitational force created by ΔM at (x, y, z) on a mass m at origin



$$|\vec{F}| = \frac{G \cdot \Delta M \cdot m}{\rho^2} \cdot \frac{\vec{\rho}}{\rho} = \frac{G \cdot \Delta M \cdot m}{\rho^3} \langle x, y, z \rangle$$

$\vec{F} = \frac{G \Delta M m}{\rho^3} \langle x, y, z \rangle$ for 1 piece, to get the total force, we have to integrate on the whole volume of the object attracting us.

So, integrating: ($\Delta M = \delta \Delta V$)

$$\vec{F} = \iiint \frac{G m \langle x, y, z \rangle \delta dV}{\rho^3}$$

Setup: • place solid so z -axis is an axis of symmetry



Then $\vec{F} = \langle 0, 0, F \rangle$

z -component: $G m \iiint \frac{z}{\rho^3} \delta dV$

* Use spherical coords:

$$G m \iiint \frac{\rho \cos \phi}{\rho^3} \delta \cdot \rho^2 \sin \phi d\rho d\phi d\theta$$

$$S_z \text{ } z\text{-component of } \vec{F} = G m \iiint \delta \cos \phi \sin \phi d\rho d\phi d\theta$$

Example: Newton's Theorem: grav. attraction of a spherical planet with uniform density is equal to that of a point mass (with same total mass) at its center